Lecture plan · for the lecturer

Loss, descent and backprop

University, 2

Time
45 min

Materials
A phone with Darsi; Paper for one hand-worked step; The slides from the kit

Learning goals

- Loss is large when wrong
- Step against the slope

Activity: In the lab

1. Run Good
2. Run Too big

Discussion

- Where does the gradient point?

Exit ticket

- θ^2 at 3, rate 0.1: write the next θ.

Try it in your next lecture
Bring three loss curves to the seminar with the rates removed. Students name the rate.

Seminar sheet · for students

Loss, descent and backprop

University, 2

Name _____ Class _____ Date _____

1 **Loss θ^2, $\theta = 3$, $\eta = 0.1$. θ after one step?**

◯ 2.4 ◯ 2.7 ◯ 3.6

2 **The gradient points at the minimum.**

◯ True ◯ False

3 **Match the curve to the cause.**

Loss grows every step	• • • •	Memorising the training set
Loss falls very slowly	• • • •	Step too small
Train loss low, new data worse	• • • •	Step too big

4 **Order one descent step.**

☐ Step against the gradient

☐ Compute the gradient

☐ Compute the loss

✎ **What will you mark if the arithmetic is right and the sentence says the gradient points at the minimum?**